

Cobham-Free Full General Automatic Rigidity for Collatz-Invariant Sequences

Removing the Final External Cobham Dependency in the ARC Program

Muteki
Independent Researcher, Japan

First released: September 23, 2026

Abstract

We study finite-kernel automatic colorings of the nonnegative integers that are invariant under the shortcut Collatz map

$$T(n) = \begin{cases} n/2, & n \equiv 0 \pmod{2}, \\ (3n+1)/2, & n \equiv 1 \pmod{2}. \end{cases}$$

Earlier stages of the ARC (Automatic Rigidity for Collatz) program established positive-domain rigidity for several individual bases, then for every odd base, and finally for all bases under an explicit Cobham-theorem handoff localized to the base-2 / pure-power-of-two branch. The present paper removes that final external dependency.

The key new step is a Cobham-free reconstruction of ARC2. For a base-2 finite-kernel automatic invariant coloring a , we pass to the odd subsequence $b(n) = a(2n+1)$. Existing ARC2 kernel arithmetic makes b base-3 finite-kernel automatic. Instead of applying Cobham's theorem to obtain ultimate periodicity, local-density machinery developed later in the ARC program shows that the adjacent-change set of b is dyadically nowhere thick. The General Odd ARC topological-pumping theorem then forces that change set to be finite. Hence b is eventually constant, and the recurrence $b(4n+2) = b(n)$ propagates the tail value to every index. Doubling invariance then recovers constancy of a on all positive integers.

Combining this new ARC2 theorem with odd-base rigidity and base-factor descent yields a single theorem for every base $B \geq 2$, with no explicit `ARCCobhamPrinciple` hypothesis and no `[Finite alpha]` typeclass assumption. The Lean 4 formalization was independently re-derived in a final audit; the final dependency report contains only `propext`, `Classical.choice`, and `Quot.sound`, with no `sorryAx`. The integrated project completed `lake build` successfully with 8906 jobs.

1 Introduction

Automatic sequences provide a natural finite-state lens through which to study arithmetic dynamics. The ARC program asks a deliberately restricted question: if a finite-state coloring is invariant along every edge of the shortcut Collatz graph, how much freedom can remain?

Let

$$T(2n) = n, \quad T(2n+1) = 3n+2,$$

which is equivalent on odd inputs to $(3m+1)/2$. A coloring $a : \mathbb{N}_0 \rightarrow \alpha$ is called *shortcut-Collatz invariant* if

$$a(T(n)) = a(n) \quad (n \in \mathbb{N}_0).$$

The ARC objective is not to prove the Collatz conjecture. Rather, it is to show that a broad class of finite-state invariants is too rigid to distinguish positive integers.

The earlier development proceeded through increasingly general cases. ARC2 and ARC3 used kernel transfer together with an explicit external form of Cobham's theorem. ARC5 and ARC7 introduced local synchronization, dyadic thinness, pumping, and finite-change arguments. ARC15 stress-tested the method at an odd composite base. These ideas were then generalized to every odd base. Base-factor descent subsequently reduced every mixed even base $B = 2^s m$ with odd $m > 1$ to its odd part.

At that point, the only remaining explicit Cobham dependency in the integrated all-base theorem was the pure power-of-two branch $B = 2^s$, because factor descent sends this branch to base 2, and the old ARC2 endpoint still passed through Cobham.

The purpose of the present paper is to close precisely that gap.

Theorem 1.1 (Cobham-free Full General ARC). *Let $B \geq 2$ and let $a : \mathbb{N}_0 \rightarrow \alpha$ have finite base- B kernel. Assume*

$$a(T(n)) = a(n) \quad (n \in \mathbb{N}_0).$$

Then there exists $c \in \alpha$ such that

$$a(n) = c \quad (n > 0).$$

The value at zero is not determined; this is sharp.

The proof has three layers:

- (i) reconstruct ARC2 without Cobham;
- (ii) use that reconstruction to close every pure power-of-two base;
- (iii) combine the pure-power branch with the already Cobham-free odd and mixed even branches.

2 Finite-kernel automaticity and shortcut invariance

For a base $b \geq 2$, define the b -kernel of a sequence $a : \mathbb{N}_0 \rightarrow \alpha$ by

$$\mathcal{K}_b(a) = \{n \mapsto a(b^e n + r) : e \geq 0, 0 \leq r < b^e\}.$$

We use the finite-kernel condition

$$\mathcal{K}_b(a) \text{ is finite}$$

as the formal notion of base- b automaticity. This is the interface used by the principal Lean theorems.

Shortcut invariance immediately yields the branch identities

$$a(2n) = a(n), \quad a(2n + 1) = a(3n + 2). \tag{1}$$

The first identity is especially important: for every $k \geq 0$,

$$a(2^k n) = a(n). \tag{2}$$

Equation (2) is the source of both factor descent and the final propagation of eventual constancy to all positive indices.

3 Where the old proof still used Cobham

Suppose the base is even. Write

$$B = 2^s m, \quad s > 0, \quad m \text{ odd.}$$

The ARC base-factor descent theorem shows that, under doubling invariance,

$$(2^s m)\text{-finite-kernel automatic} \implies m\text{-finite-kernel automatic.} \quad (3)$$

If $m > 1$, then $m \geq 3$ is odd. The completed General Odd ARC theorem already gives positive-domain constancy directly, so this branch is Cobham-free.

The only unresolved case after localization was

$$m = 1, \quad \text{i.e.} \quad B = 2^s.$$

Applying (3) repeatedly reduces this branch to base 2. Thus the all-base proof becomes Cobham-free as soon as ARC2 itself is reconstructed without Cobham.

This observation is conceptually useful: the problem is not “remove Cobham everywhere.” By the time of the present work, the dependency had already been localized to one sharply identified endpoint.

4 Cobham-free reconstruction of ARC2

Let $a : \mathbb{N}_0 \rightarrow \alpha$ be base-2 finite-kernel automatic and shortcut-Collatz invariant. Define the odd subsequence

$$b(n) := a(2n + 1). \quad (4)$$

The new proof does not try to prove that b is shortcut-Collatz invariant. Instead, it extracts exactly the three properties needed for rigidity: base-3 finite-kernel automaticity, dyadic nowhere-thickness of the adjacent change set, and the self-similarity relation $b(4n + 2) = b(n)$.

4.1 The odd subsequence is base 3 automatic

The original ARC2 arithmetic already contains the essential kernel transfer. Using (1), every relevant base-3 decimation of the odd subsequence can be represented inside the finite base-2 kernel generated by a . In particular,

$$\mathcal{K}_3(b)$$

is finite.

Formally, the old lower-level theorem `arc_three_kernel_odd_finite` is reused directly. The new wrapper

$$\text{arc2OddSubseq_threeAutomatic_of_twoAutomatic}$$

works purely at finite-kernel level and avoids the unnecessary `[Finite alpha]` hypothesis carried by an older high-level interface.

Proposition 4.1. *If a has finite base-2 kernel and satisfies shortcut invariance, then $b(n) = a(2n+1)$ has finite base-3 kernel.*

This is an important point in the formal architecture: no new arithmetic kernel-transfer theorem was needed. What changed was the way the existing transfer theorem was connected to the later topological machinery.

4.2 The adjacent-change set is dyadically nowhere thick

Define

$$\mathcal{D}_b = \{n \in \mathbb{N}_0 : b(n) \neq b(n+1)\}.$$

Since

$$b(n) = a(2n+1), \quad b(n+1) = a(2n+3),$$

the equality $b(n) = b(n+1)$ is equivalent to

$$a(2n+1) = a(2n+3). \tag{5}$$

The decisive new observation is that the local-density theorem for shortcut-invariant colorings can be applied with gap $H = 2$.

Informally, fix any dyadic parent cylinder

$$n \equiv r \pmod{2^m}.$$

Under the affine map $x = 2n+1$, this becomes the odd dyadic cylinder

$$x \equiv 2r+1 \pmod{2^{m+1}}.$$

The local-density theorem supplies a deeper subcylinder on which

$$a(x) = a(x+2).$$

Pulling this subcylinder back through $x = 2n+1$ gives a deeper dyadic child in the original n -variable on which

$$b(n) = b(n+1).$$

Therefore that child contains no element of $\mathcal{D}_b$.

Proposition 4.2 (Dyadic thinness). *For every dyadic parent cylinder there is a deeper dyadic child disjoint from $\mathcal{D}_b$. Equivalently, $\mathcal{D}_b$ is dyadically nowhere thick.*

In Lean this is established through

`arc2OddSubseq_local_gap_two_child,` `arc2OddSubseq_changeSet_nowhereThick,`

and packaged as

`arc2CobhamRemoval_stage1_thinness.`

This is the step that replaces the role previously played by Cobham's theorem. The old proof obtained global regularity through ultimate periodicity. The new proof obtains a different form of global control: automaticity plus dyadic topological thinness forces only finitely many adjacent changes.

4.3 Finite change set and eventual constancy

The General Odd ARC development proves the required topological-pumping principle for odd automatic bases. Applied at base 3, it says in the present setting that a base-3 finite-kernel automatic adjacent-change set that is dyadically nowhere thick must be finite.

Combining Proposition 4.1 and Proposition 4.2 gives

$$\mathcal{D}_b \text{ finite.}$$

Hence there exists N such that

$$b(n) = b(n+1) \quad (n \geq N),$$

and therefore b is eventually constant.

The relevant reusable General Odd ARC ingredients are the change-set automaticity, nowhere-thick-to-finite support theorem, and finite-change-to-tail constancy theorem.

4.4 Tail constancy propagates to every index

Shortcut invariance also gives the exact recurrence

$$b(4n+2) = b(n). \tag{6}$$

Indeed,

$$b(4n+2) = a(8n+5),$$

and repeated use of the shortcut branch relations returns the value to $a(2n+1) = b(n)$.

Now iterate the affine map

$$\Phi(n) = 4n+2.$$

For every fixed n , the values

$$n, \Phi(n), \Phi^2(n), \dots$$

grow without bound, while (6) preserves the color. Thus, once an iterate lies beyond the eventual-constancy threshold,

$$b(n) = c.$$

So b is constant on all of $\mathbb{N}_0$.

Finally, every positive integer can be written as

$$n = 2^k q$$

with q odd. By (2),

$$a(n) = a(q),$$

and $a(q)$ is a value of the odd subsequence b . Hence a is constant on all positive integers.

Theorem 4.3 (Cobham-free ARC2). *Let $a : \mathbb{N}_0 \rightarrow \alpha$ have finite base-2 kernel and satisfy shortcut invariance. Then there exists $c \in \alpha$ such that*

$$a(n) = c \quad (n > 0).$$

The principal Lean endpoint is

`arc2_main_kernel_form_no_cobham,`

with stage wrapper

`arc2CobhamRemoval_stage2_final.`

5 Closing every base

With Theorem 4.3 available, the last missing branch of the general argument becomes elementary.

Proposition 5.1 (Pure powers of two). *Let*

$$B = 2^s, \quad s > 0.$$

If a is base- B finite-kernel automatic and shortcut-Collatz invariant, then a is constant on the positive integers.

Proof. Write $s = t + 1$. Base-factor descent gives base-2 finite-kernel automaticity. Apply Theorem 4.3. $\square$

In Lean, this is separated into

`arc_powTwoSucc_final_no_cobham`

and

`arc_powTwoBase_final_no_cobham.`

Proposition 5.2 (Mixed even bases). *Let*

$$B = 2^s m, \quad s > 0, \quad m > 1 \text{ odd}.$$

If a is base- B finite-kernel automatic and shortcut-Collatz invariant, then a is constant on the positive integers.

Proof. Factor descent gives finite base- m kernel. Since $m \geq 3$ is odd, apply the General Odd ARC theorem. $\square$

Proof of Theorem 1.1. If B is odd, then $B \geq 3$ and General Odd ARC applies directly.

If B is even, factor

$$B = 2^s m$$

with $s > 0$ and m odd. If $m = 1$, use Proposition 5.1. If $m > 1$, use Proposition 5.2. These cases exhaust all $B \geq 2$. $\square$

The packaged Lean endpoint is

`arc_full_general_no_cobham.`

6 Lean 4 formalization

The Cobham-removal layer was developed conservatively: each stage was checked standalone, integrated into the root import graph, and followed by a project build before proceeding.

File	Role	Principal endpoint
ARC2CobhamRemoval Stage0.lean	Odd-subsequence packaging, recurrence extraction, eventual-to-global propagation, and recovery of positive-domain constancy.	arc2OddSubseq_rigidity_of_threeAutomatic_and_thin
ARC2CobhamRemoval Stage1.Thinness_v3.lean	Derives dyadic nowhere-thickness of the odd-subsequence adjacent-change set from shortcut invariance and local density.	arc2CobhamRemoval_stage1_thinness
ARC2CobhamRemoval Stage2.Final_v2.lean	Reconnects directly to the lower ARC2 finite-kernel transfer theorem and closes ARC2 without Cobham or <code>[Finite alpha]</code> .	arc2CobhamRemoval_stage2_final
ARC2CobhamRemoval Stage3.FullGeneral.lean	Replaces the pure power-of-two branch, reassembles all even bases, and closes every base.	arc_full_general_no_cobham
ARCFullGeneralNoCobham FinalAudit.lean	Independently reconstructs the all-base theorem from lower-level components and runs representative smoke tests.	arcFinalAudit_full_general_no_cobham

6.1 Final audit

The final audit deliberately does not prove the all-base result by merely calling the packaged Stage-3 endpoint. It reconstructs the three structural branches:

pure 2^s , mixed even, odd,

from lower-level theorems.

Representative concrete instances were checked at bases

2, 3, 4, 5, 6, 7, 10, 15, 64, 100.

For the principal final theorems, `\#print axioms` reports

`propext`, `Classical.choice`, `Quot.sound`,

and does not report `sorryAx`. No explicit `ARCCobhamPrinciple` parameter occurs in the final theorem, and no `[Finite alpha]` typeclass assumption is required.

After the final audit import was added to the root project, the command

`lake build`

completed successfully with **8906 jobs**.

Remark 6.1 (Scope of the formal audit). The Lean dependency report verifies the proof as encoded relative to the imported definitions and Mathlib environment. It is not, by itself, an independent peer review of the mathematical modeling choices or a literature novelty certificate.

7 Relation to the earlier ARC papers

The present release is intentionally additive rather than substitutive.

The earlier ARC papers are retained because they document the route by which the final architecture emerged:

- ARC2 contains the original base-2 kernel arithmetic and the first rigidity route.
- ARC3 develops a complementary kernel-transfer case.

- ARC5 introduces local synchronization, dyadic thinness, and a finite-change strategy.
- ARC7 develops doubled-loop pumping and dyadic filling.
- ARC15 stress-tests the mechanism at an odd composite base.
- General Odd ARC extracts the common odd-base structure.
- The all-even and first Full General integrations identify exactly where the external Cobham handoff remained.

The new proof feeds machinery developed later in the program back into ARC2. That feedback loop is the conceptual content of the present paper.

Thus this work should not be described as correcting an error in the earlier papers. The earlier theorems remain valid under their stated proof architectures. The present result removes an external hypothesis from the integrated formal route and therefore strengthens the final theorem.

8 Sharpness at zero

The restriction to positive integers cannot be removed.

Proposition 8.1. *Let $x, y \in \alpha$ be arbitrary and define*

$$a(0) = x, \quad a(n) = y \quad (n > 0).$$

Then a is finite-kernel automatic in every base and shortcut-Collatz invariant.

Consequently, the value at zero is genuinely free. Theorem 1.1 is sharp in its domain of constancy.

For Boolean colorings, Theorem 1.1 may equivalently be read as a rigidity statement for automatic invariant subsets of $\mathbb{N}_0$: on the positive integers, such a set is either empty or all of $\mathbb{N}_{>0}$, while membership of zero remains independent.

9 What the theorem does not say

It is important to separate the ARC theorem from the Collatz conjecture itself.

Theorem 1.1 says that a finite-kernel automatic coloring cannot provide a nontrivial positive-domain invariant of the shortcut Collatz map. It does not assert that every Collatz orbit reaches 1, nor does it rule out all conceivable invariants. In particular, arbitrary nonautomatic or infinite-state colorings lie outside the theorem's hypotheses.

Similarly, removing Cobham from the present proof does not diminish the importance of Cobham's theorem in automatic-sequence theory. The result is specific to the additional arithmetic and dynamical structure supplied by shortcut-Collatz invariance.

10 Release note

First released: September 23, 2026.

Earlier ARC papers are not superseded or withdrawn. They are retained as historical stages of the development. The present paper is added as a new release documenting the Cobham-free closure of the Full General ARC theorem.

Acknowledgment of AI assistance

OpenAI ChatGPT (GPT-5.6 Sol) was used as a research and drafting aid for mathematical exploration, theorem organization, Lean 4 code development and debugging, formal dependency auditing, manuscript preparation, and repository organization. The AI system is not an author or bibliographic source. Responsibility for the released mathematical claims, formalization, references, and wording remains with the human author.

References

- [1] J.-P. Allouche and J. Shallit, *Automatic Sequences: Theory, Applications, Generalizations*, Cambridge University Press, 2003.
- [2] A. Cobham, On the base-dependence of sets of numbers recognizable by finite automata, *Mathematical Systems Theory* **3** (1969), 186–192.
- [3] S. Eilenberg, *Automata, Languages, and Machines, Vol. A*, Academic Press, 1974.